International AS · Edexcel · Chemistry

7 mins   7 questions

Multiple Choice Questions

Organic Chemistry: Techniques & Spectra

Organic Chemistry Techniques / Interpreting Mass Spectra / Using Infrared Spectrometry

Easy (5 questions)	/5
Medium (2 questions)	/2
Total Marks	**/7**

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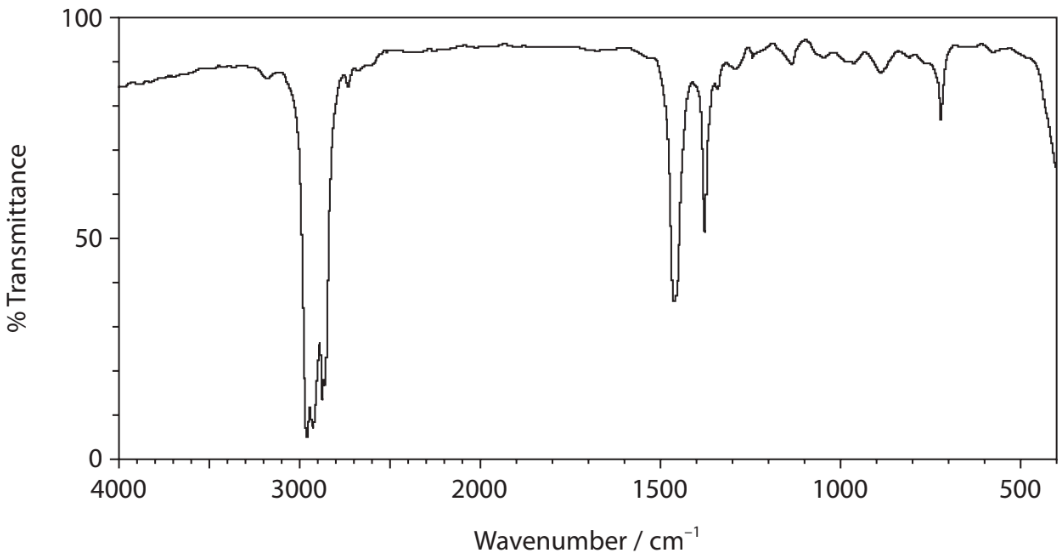
Easy Questions

1 The infrared spectrum of an organic compound is shown.



Which of these compounds would give this infrared spectrum?

A. hexanal

B. hexane

C. hexanoic acid

D. hexan-1-ol

(1 mark)

2 In a mass spectrum, the molecular ion is the ion which always has the

A. greatest abundance

B. greatest stability

C. highest charge

D. highest mass/charge ratio

(1 mark)

3 Butan-1-ol and butan-2-ol are isomers.

Which *m / z* value would be expected to have a significant peak in the mass spectrum of butan-1-ol but **not** that of butan-2-ol?

A. 15

B. 29

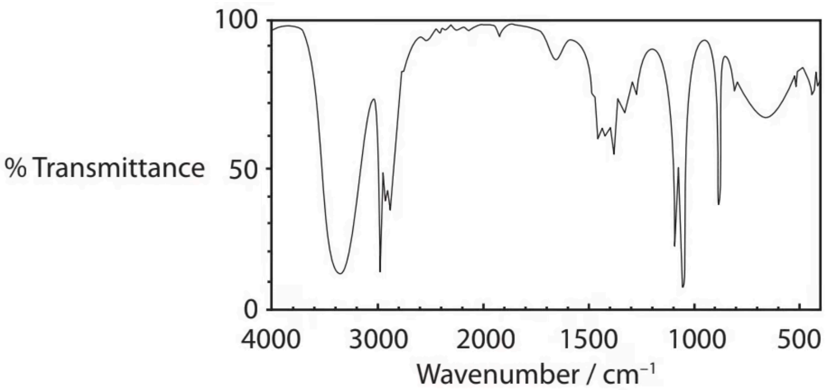
C. 43

D. 57

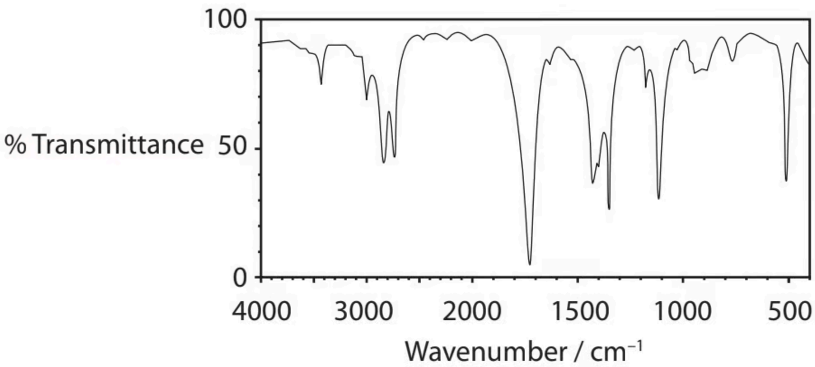
(1 mark)

4 Which could be the infrared spectrum of $CH_2{=}CHCH_2OH$?

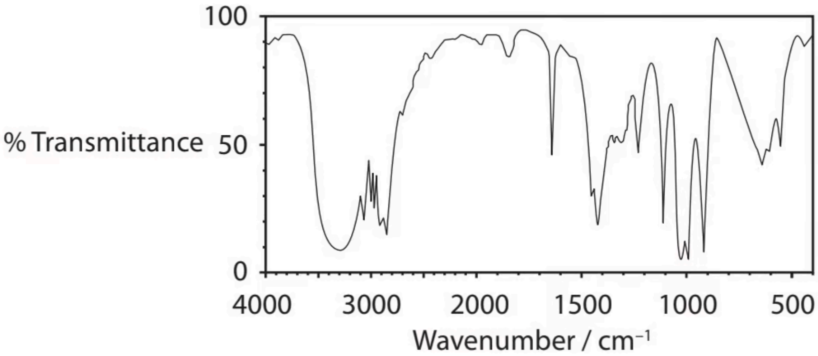
☐ **A**



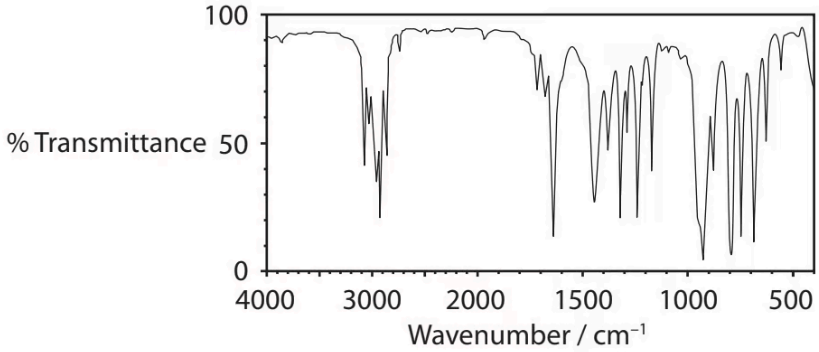
☐ **B**



C

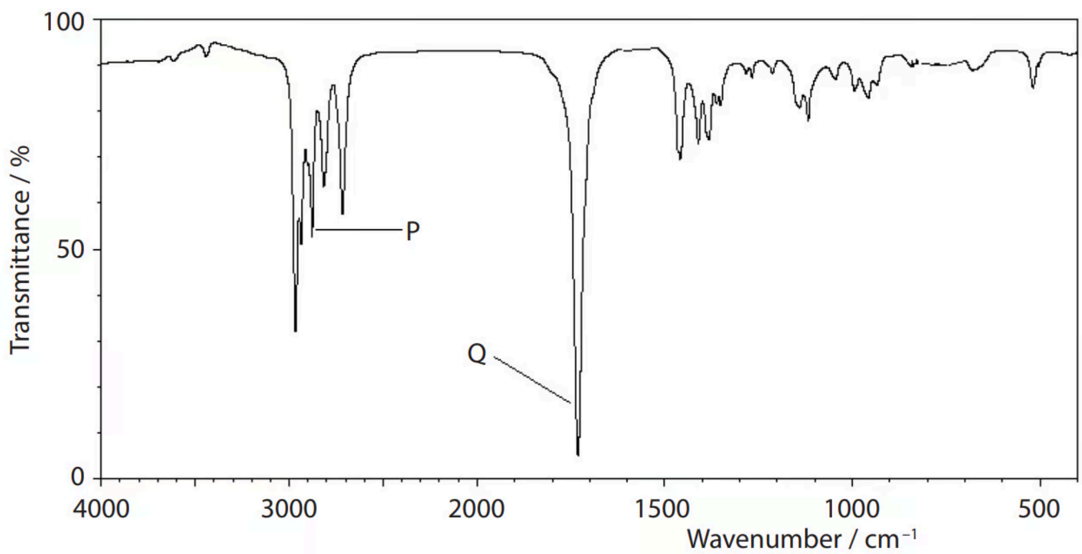


D



(1 mark)

5 An aliphatic organic compound has the infrared spectrum shown.



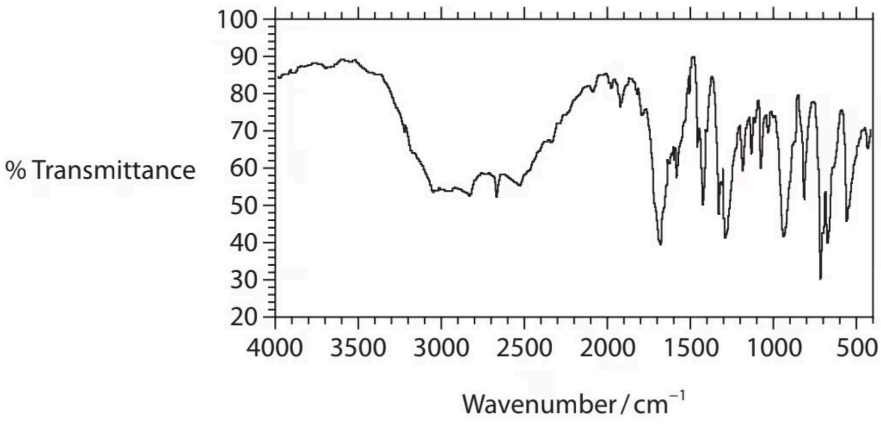
What are the bond stretches responsible for the peaks **P** and **Q** in the spectrum?

		P	Q
☐	**A**	O—H carboxylic acid	C═O carboxylic acid
☐	**B**	O—H carboxylic acid	C═O aldehyde
☐	**C**	C—H aldehyde	C═O carboxylic acid
☐	**D**	C—H aldehyde	C═O aldehyde

(1 mark)

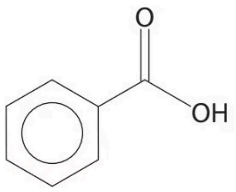
Medium Questions

1 The infrared spectrum of a compound **X** is shown.

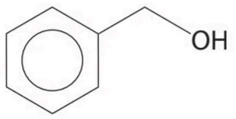


Which could be compound **X**?

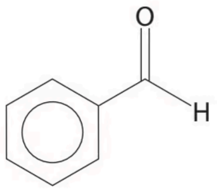
☒ **A**



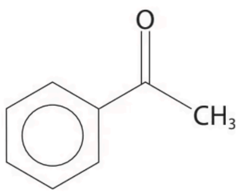
☒ **B**



☒ **C**

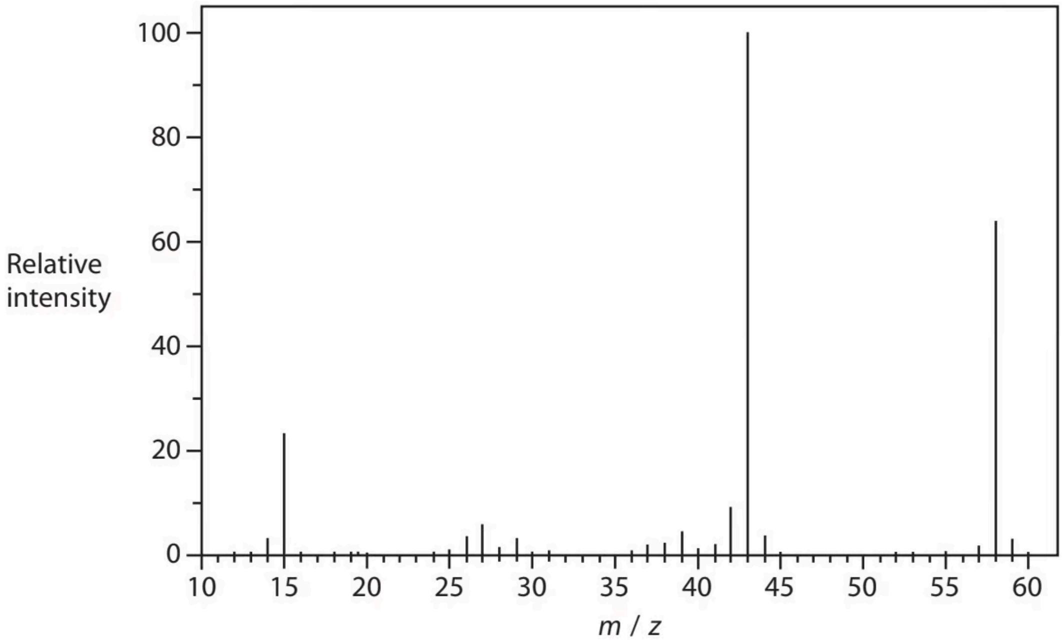


☒ **D**



(1 mark)

2 The mass spectrum of propanone is shown.



Which fragment is most likely to produce the peak at *m* / *z* = 43?

A. $CH_3CH_2CH_2^+$

B. CH_3CO^+

C. CH_2CHO^+

D. $CHCH_2O^+$

(1 mark)